CSA1706-ARTIFICIAL INTELLIGENCE

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1. Smart Home Automation System Using Intelligent Agents

Introduction

A **Smart Home Automation System** is an Artificial Intelligence (AI) application that automatically controls home devices such as **lighting, temperature, and security**. The system continuously monitors the environment through sensors (motion detectors, temperature sensors, light sensors, cameras, and time-based inputs) and makes intelligent decisions to improve **comfort, energy efficiency, and security**. Different types of intelligent agents can be used to perform these tasks.

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1. Simple Reflex Agent

Definition

A **Simple Reflex Agent** selects actions based only on the **current percept** (current sensor input). It uses predefined **condition–action rules** and does not remember previous states.

Working in the Smart Home

Examples:

- If motion is detected in a room → Turn ON the lights.
- If no motion is detected for 10 minutes → Turn OFF the lights.
- If smoke is detected → Activate the fire alarm.

Advantages

- Simple and easy to implement.
- Fast response to sensor inputs.
- Suitable for basic automation tasks.

Disadvantages

- Cannot remember previous events.
- Cannot learn user preferences.
- May not perform well in changing environments.

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2. Model-Based Reflex Agent

Definition

A **Model-Based Reflex Agent** maintains an **internal model (memory)** of the environment. It combines current sensor information with previous observations to make better decisions.

Working in the Smart Home

Examples:

- Remembers whether windows are open before switching on the air conditioner.
- Keeps track of room occupancy.
- Detects whether someone has recently entered or left the house.

Advantages

- Can handle partially observable environments.
- Makes more intelligent decisions than a simple reflex agent.
- Maintains information about previous states.

Disadvantages

- More complex than a simple reflex agent.
- Requires memory and additional processing.

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3. Goal-Based Agent

Definition

A **Goal-Based Agent** selects actions that help achieve a specified goal.

Working in the Smart Home

Possible goals:

- Maintain room temperature at **24°C**.
- Ensure all doors are locked before midnight.
- Minimize electricity consumption.
- Maintain user comfort throughout the day.

The agent evaluates different actions and chooses the one that best satisfies the desired goal.

Advantages

- Can plan future actions.
- Produces intelligent and goal-oriented behavior.
- Flexible in handling different situations.

Disadvantages

- Requires search and planning.
- Decision-making may take more time.

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4. Utility-Based Agent

Definition

A **Utility-Based Agent** chooses the action that provides the **maximum overall benefit (utility)**. It evaluates multiple factors simultaneously and selects the best possible action.

Working in the Smart Home

Examples:

- Balances comfort and electricity usage.
- Chooses the best air conditioner setting to reduce power consumption while maintaining a comfortable temperature.
- Decides whether to activate outdoor security lights based on motion detection, time of day, and energy usage.

Advantages

- Produces the best overall decision.
- Handles conflicting objectives effectively.
- Maximizes user satisfaction and energy efficiency.

Disadvantages

- More computationally expensive.
- Requires a utility function.

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Simple Reflex	No	No	No	Basic automation

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Model-Based	Yes	No	Limited	Environment monitoring
Goal-Based	Yes	Yes	Limited	Planning and automation
Utility-Based	Yes	Yes	Yes	Comfort, security, and energy optimization

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Best Approach

The most effective solution is a **Hybrid Approach** combining:

- **Model-Based Agent** – remembers previous states and user behavior.
- **Goal-Based Agent** – achieves objectives such as maintaining temperature and ensuring security.
- **Utility-Based Agent** – optimizes comfort, energy efficiency, and safety simultaneously.

Justification

A hybrid approach is the best choice because it:

- Learns user preferences over time.
- Responds to real-time sensor inputs.
- Balances comfort and energy consumption.
- Improves home security.
- Adapts to changing environmental conditions.

Thus, a **Model-Based + Goal-Based + Utility-Based Hybrid Agent** provides the most intelligent, flexible, and efficient smart home automation system.

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2. Robot Navigation in an Unknown Maze

Introduction

A robot is placed in an **unknown maze** and must find a path from the **Start (S)** position to the **Goal (Exit)** without any prior knowledge of the maze layout.

Since the robot does not know the environment, it must use **uninformed search strategies** to explore and eventually find the exit.

Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the **lowest cumulative path cost** first. It guarantees the least-cost solution when all costs are non-negative.

Working

1. Start from the initial position.
2. Insert the start node into a priority queue.
3. Expand the node with the lowest path cost.
4. Add neighboring nodes with updated costs.
5. Continue until the goal is reached.

Advantages

- Finds the optimal (least-cost) path.
- Works with varying movement costs.
- Complete and optimal.

Disadvantages

- May expand many unnecessary nodes.
- High memory usage due to storing all frontier nodes.
- Slower for large search spaces.

Complexity

- **Time Complexity:** $O\left(b^{1+\lfloor C^*/\epsilon \rfloor}\right)$
- **Space Complexity:** Same as time complexity (stores all generated nodes).

Where:

- b= branching factor
- C^*= optimal solution cost
- ϵ= minimum step cost

Iterative Deepening Search (IDS)

Definition

Iterative Deepening Search repeatedly performs **Depth-First Search (DFS)** with increasing depth limits until the goal is found.

Working

1. Perform DFS with depth limit = 0.
2. If the goal is not found, increase the depth limit to 1.
3. Repeat with limits 2, 3, and so on until the goal is reached.

Advantages

- Requires very little memory.
- Complete for finite search spaces.
- Suitable when the solution depth is unknown.

Disadvantages

- Repeats exploration of upper-level nodes.
- Less efficient than UCS when costs differ.
- Not suitable for varying path costs.

Complexity

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(bd)$

Where:

- b= branching factor
- d= depth of the solution

Comparison of UCS and IDS

Feature	**Uniform Cost Search**	**Iterative Deepening Search**
Search Strategy	Lowest path cost first	Increasing depth limits
Completeness	Yes	Yes
Optimality	Yes (for non-negative costs)	Yes (when step costs are equal)
Time Complexity	High	Moderate
Space Complexity	High	Low
Best Used For	Variable-cost paths	Unknown-depth problems

Relation to Intelligent Agents

The maze-solving robot acts as an **intelligent agent** because it:

- **Perceives** the environment using sensors.
- **Processes** the information it receives.
- **Chooses** actions that move it toward the goal.
- **Acts** by moving through the maze.

Depending on its capabilities, the robot may behave as:

- **Simple Reflex Agent:** Moves only based on current surroundings; unsuitable for unknown mazes.
- **Model-Based Agent:** Builds an internal map while exploring; avoids revisiting explored paths.
- **Goal-Based Agent:** Plans actions to reach the exit.
- **Utility-Based Agent:** Chooses paths that minimize travel cost, time, or risk.

Most Effective Combination

The most suitable combination is:

- **Model-Based Goal-Based Agent**
- **Uniform Cost Search (UCS)**

Justification

This combination is effective because:

- The **Model-Based Agent** stores information about explored areas, enabling navigation in an unknown environment.
- The **Goal-Based Agent** keeps the robot focused on reaching the exit.
- **Uniform Cost Search** guarantees the least-cost path when movement costs vary.
- The robot can continuously update its internal map and make informed decisions.

If all movement costs are equal and memory is limited, **Model-Based Agent + Iterative Deepening Search (IDS)** is also a practical solution due to its lower memory requirements.

Conclusion

- A **Hybrid Intelligent Agent (Model-Based + Goal-Based + Utility-Based)** is the best choice for a smart home because it ensures **comfort, energy efficiency, adaptability, and security**.
- For robot navigation in an unknown maze, a **Model-Based Goal-Based Agent using Uniform Cost Search** is the most effective solution, as it learns the environment, plans efficiently, and guarantees an optimal path under varying costs.